

Lesson 1:

Statistics Overview



Goals

- To understand introduction to statistics.
- To understand what descriptive statistics are used for.
- To understand the differences between the 3 types of Univariate Analysis: Central Tendency, Dispersion and Frequency Distributions.
- To know the basic Excel functions that can be used to determine descriptive statistics.

What is statistic?

- A statistic is a number that represents a property of a sample

Population = parameter

Sample = statistic

Singular Sense	Plural Sense
In this sense, 'Statistics' refers to what is called statistical methods, which deal with the collection, classification, tabulation, analysis and interpretations of statistical data. So, it is described as a branch of the science which deals with numerical facts and make decisions as well. Every the statistical process should pass through these stages.	In the plural sense, 'Statistics' means a set of numerical information known as a statistical data, i.e., information on population features, climatic variables, enrolment of students in a class, and so on.

What is statistic?

1. Collecting Data

– e.g. Survey

2. Presenting Data

– e.g., Charts & Tables

3. Characterizing Data

– e.g., Average

**Data
Analysis**

Why?



**Decision-
Making**



Origin and Growth of Statistics

The word 'Statistics' and 'Statistical' are all derived from the Latin word Status, means a political state.

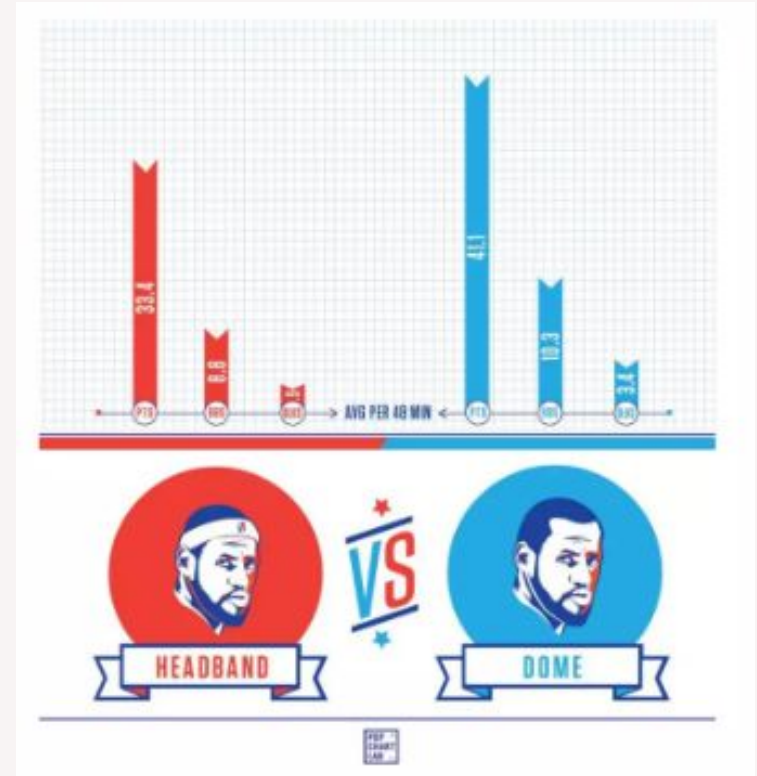
- Statistics is concerned with scientific methods for
 - ❖ Collecting
 - ❖ Organising
 - ❖ Summarising
 - ❖ Presenting and analysing data

as well as deriving valid conclusions and making reasonable decisions on the basis of this analysis.

The bottom of the slide features decorative geometric elements. On the left, there are three horizontal lines of varying lengths, each ending in a small circle. Below these, there are several overlapping triangles in shades of blue and grey, creating a modern, abstract design.

Functions of Statistics

1. Condensation
2. Comparison
3. Forecasting
4. Estimation



Scope of statistics

1. **Statistics and Industry**
2. **Statistics and Commerce**
3. **Statistics and Agriculture**
4. **Statistics and Economics**
5. **Statistics and Education**
6. **Statistics and Planning**
7. **Statistics and Medicine**
8. **Statistics and Modern applications**



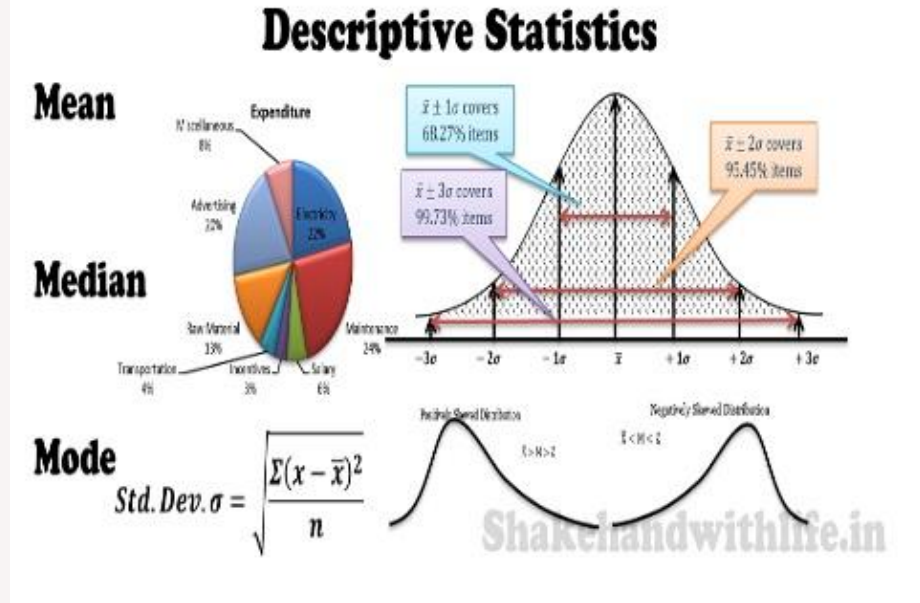
Univariate Analysis

- A technique used to analyze a single INDEPENDENT* variable or a single DEPENDENT variable
- This is the simplest way of analyzing data
- “Uni” = one
- Aim to generate a summary that describes **patterns** associated with the data.
- Not examining **relationships** or causes.
- Patterns and relationships are not the same thing!

*also known as the predictor variable

Descriptive Statistics

- Help describe the basic features of the sample data
- Provide simple summaries of the sample
- Broken down into measures of tendency (mean, median, mode) and measures of spread (Standard/Deviation, ranges, quartiles, absolute deviation, Variance)



For example: Clothing Store

- Imagine you own a clothing store, and you are keeping track of monthly sales and inventory
- What things would you want to know?
 - Important statistics would be:
 - The average (mean) sales for that month
 - The top item sold that month (mode)

Very useful for big data sets, where it would be hard to visualize data trends without descriptive statistics!

However, with descriptive statistics alone, you cannot make conclusions beyond the data analyzed – they are just used to describe the data

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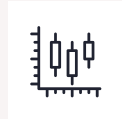
*also known as the predictor variable

Describing Data with Univariate Analysis

- 3 different ways



1- Central
Tendency



2- Dispersion



3- Frequency
Distribution

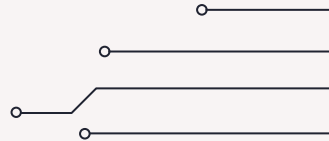
Central Tendency

Indicates where the center of the distribution tends to be

MODE

MEAN

MEDIAN



MODE

The most frequently occurring value in your data set.



11	21	38	46	16	22	34	52
19	25	38	55	20	26	39	58

38 is the only number that occurs twice, therefore it is the most frequently occurring number.

38 = MODE

MEAN

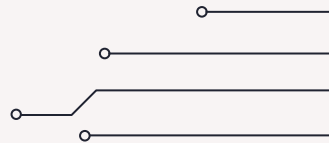
The average value in the data set

You can find the mean by summing all the numbers in the data set, and dividing by how many numbers there are

11 21 38 46 16 22 34 52
19 25 38 55 20 26 39 58

$$\begin{array}{r} 11 + 21 + 38 + 46 + 16 + 22 + 34 + 52 + 19 \\ + 25 + 38 + 55 + 20 + 26 + 39 + 58 \\ \hline 16 \end{array}$$

$$32.5 = \text{MEAN}$$



Median

The middle value in the data set



11 16 19 20 21 22 25 26 34
38 38 39 46 52 55 58

When you have a dataset with an even number of values like (like here, we have 16 data values), the median number is the average of two number which 'surround' the middle

$$\frac{26 + 34}{2}$$

30 = MEDIAN

Dispersion

Degree of variation in the data

ABSOLUTE DEVIATION AND VARIANCE

STANDARD DEVIATION

QUARTILES

RANGE

Absolute Deviation

The average 'distance' of all of the elements of the elements in data set from the mean of the same data set.

Step 1: Find the mean of the data.

Step 2: Find the absolute deviations.

Subtract the mean from each element. (Make all values positive)

Step 3: Find the mean of the absolute deviations obtained in step 2.

Why do we say distance?

Think about a simple data set: 1, 2, 3

Mean = $(1 + 2 + 3)/3 = 2$

1) $|1-2| = 1 \rightarrow$ 'distance' is same as difference here!

2) $|2-2| = 0$

3) $|3-2| = 1$

Usually say distance because it makes 'physical sense'

Example of Absolute Deviation

Example: Find the Mean Absolute Deviation of 3, 12, 7, 2

Step 1: Find the mean of the data.

$$\frac{3+12+7+2}{4} = \frac{24}{4} = 6$$

Step 2: Find the absolute deviations.

$$|3-6|=3$$

$$|12-6|=6$$

$$|7-6|=1$$

$$|2-6|=4$$

Step 3: Find the mean of the absolute deviations.

$$\frac{3+6+1+4}{4} = \frac{14}{4} = 3.5$$

Variance

A measurement of spread between numbers in a dataset.

MEAN SQUARED DEVIATION

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

- σ^2 = variance
- N = number of members in the data set
- x_i = i^{th} member
- μ = mean
- $\sum$ = summation $\rightarrow$ adding up everything!

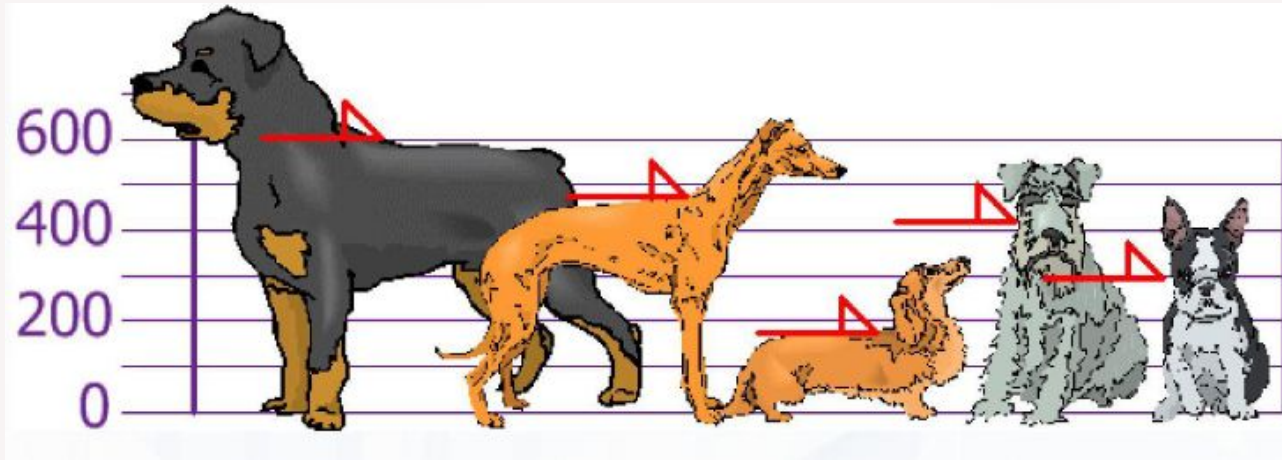
Variance = (Standard deviation)²

<https://www.mathsisfun.com/data/standard-deviation.html>

Example of Variance

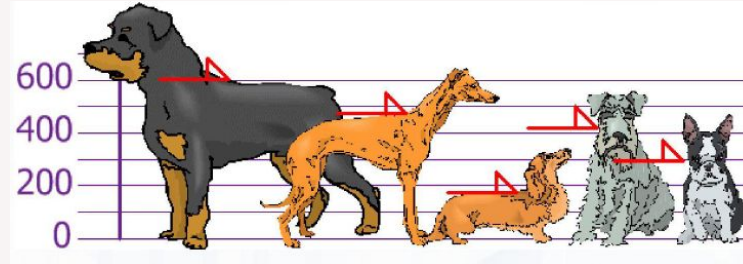
- Imagine you have measured the shoulder heights of dogs, and you want to calculate the variance!
- 600mm, 470mm, 170mm, 430mm and 300mm.

e.g. dogs



Example of Variance

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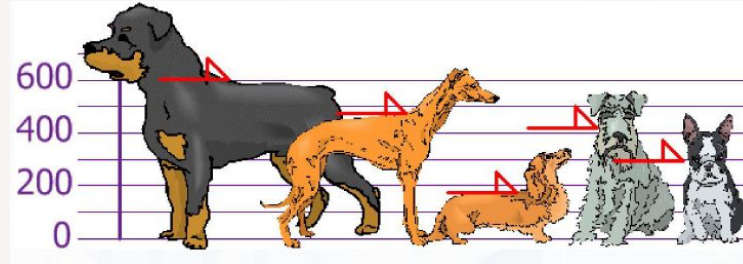
$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

First thing you need to do is calculate the mean

Mean = (600mm+ 470mm+ 170mm+ 430mm+ 300mm)/ 5 = 1970/5 = 394

Example of Variance

- Imagine you have measured the shoulder heights of dogs, and you want to calculate the variance!
- 600mm, 470mm, 170mm, 430mm and 300mm.



e.g. dogs

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Mean = 394

Squared deviation repeat for other ones

$$|600 - 394|^2 = (206)^2 = 42436$$

$$|470 - 394|^2 = (76)^2 = 5776$$

...repeat for other ones....

42436, 5776, 50176, 1296, 8936

Add all squared deviation in terms together:

$$42436 + 5776 + 50176 + 1296 + 8930 = 108620$$

Divide by the number of terms:

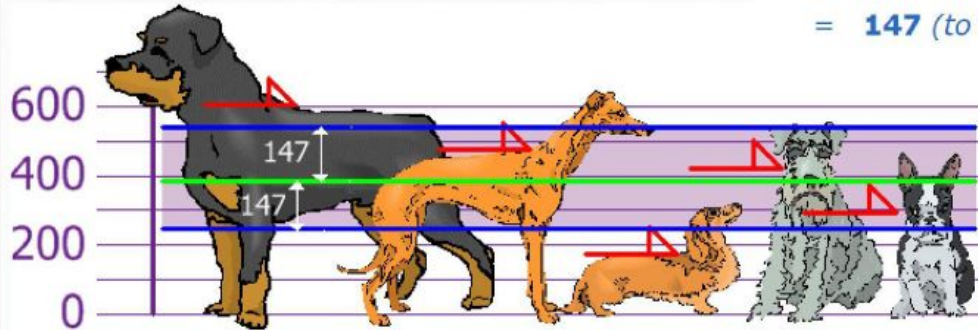
$$108620 / 5 = \text{VARIANCE}$$

Standard Deviation

Standard Deviation is the **SQUARE ROOT** of variance

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

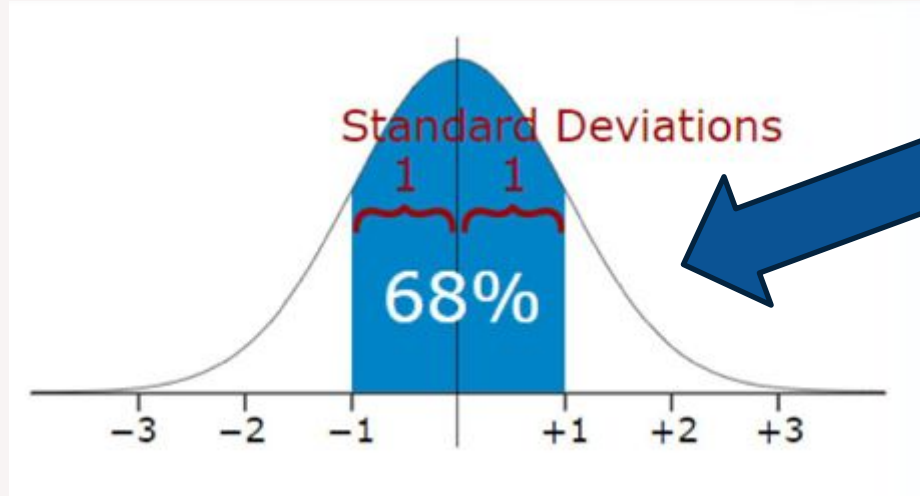
$$\begin{aligned}\sigma &= \sqrt{21704} \\ &= 147.32... \\ &= \mathbf{147} \text{ (to the nearest mm)}\end{aligned}$$



Good thing about standard deviation!

You can say which dog heights lie within one standard deviation of the mean.
Then you can say which dogs are unusually high or short.

Standard Deviation



You can expect around 68% of your data values to be within **ONE STANDARD DEVIATION OF THE MEAN**

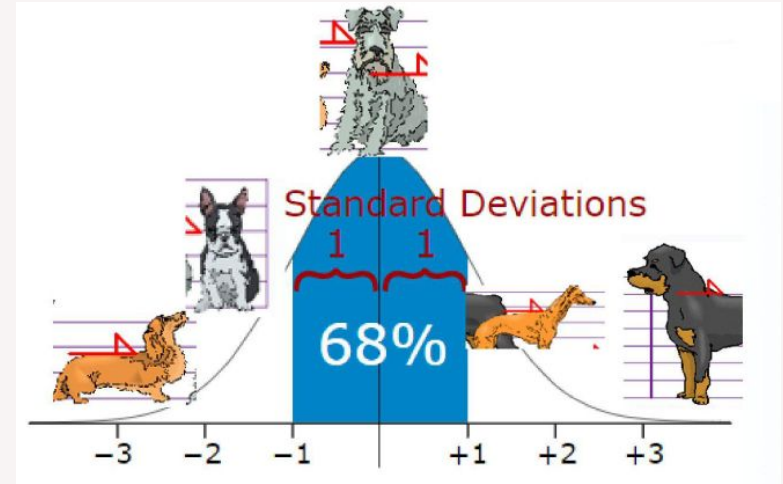
This is a **NORMAL DISTRIBUTION** curve - what does it mean?

- It shows on the x axis the difference between a data value and the mean value
- It shows on the y axis frequency
- Typically, most of your values are going to be around the mean, which is why the frequency is highest around the middle
- As you get further from the mean → data 'extremes'

Standard Deviation

Why do we use standard deviation instead of variance more often?

Because standard deviation is closer to the values of the mean, so it makes more sense to use in analysis.



Quartiles

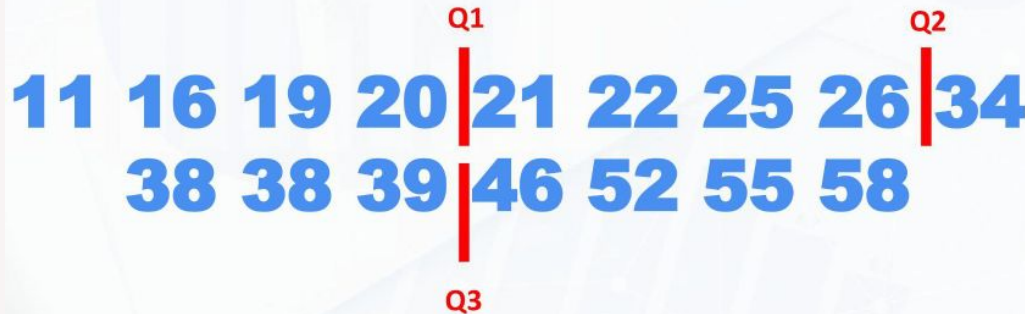
Quartiles separate the data set in to quarters, or quartiles



- **First quartile, or lower quartile (Q1):** below this point, is one quarter of the data.
- **Second quartile, or median (Q2):** below this point, is one half of the data.
- **Third quartile, or upper quartile (Q3):** below this point is three quarters of the data.

Quartiles

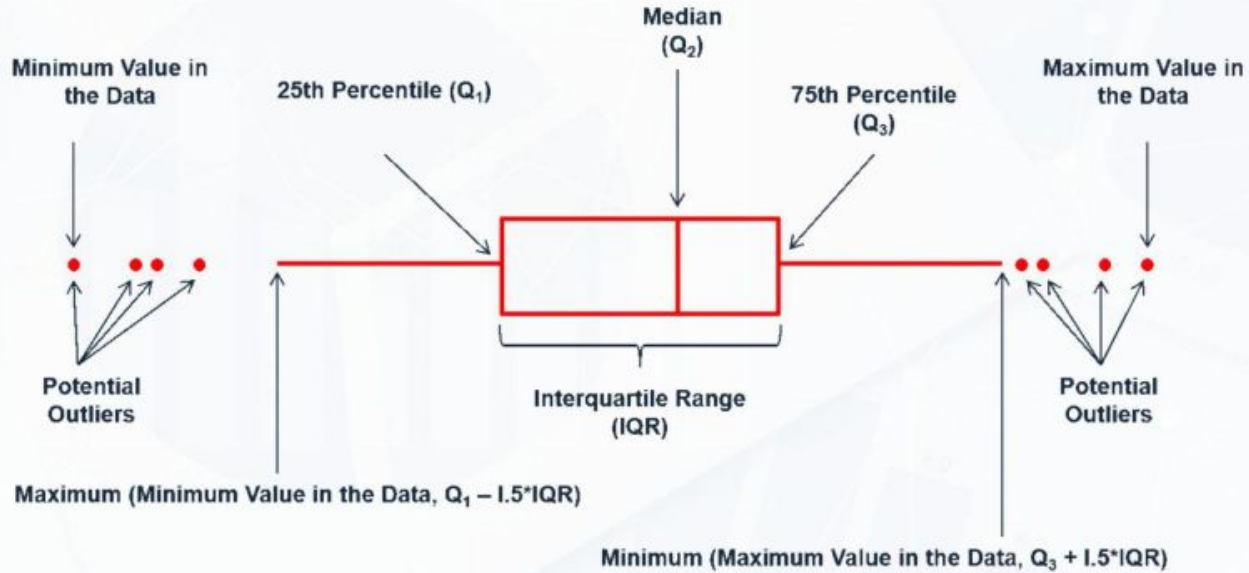
ALWAYS REMEMBER TO
ORDER YOUR DATA IN
ASCENDING ORDER FIRST !!



$$\text{Interquartile Range} = Q3 - Q1 = ((39+46)/2) - ((20+21)/2) = 21$$

The Interquartile Range is a BASIC measure of spread of the data

Quartiles



BOX PLOT

Range

Range is the difference between the largest and the smallest observation in the data.

The highest data value – lowest data value

RANGE FORMULA

$$R = \text{Maximum Value} - \text{Minimum Value}$$

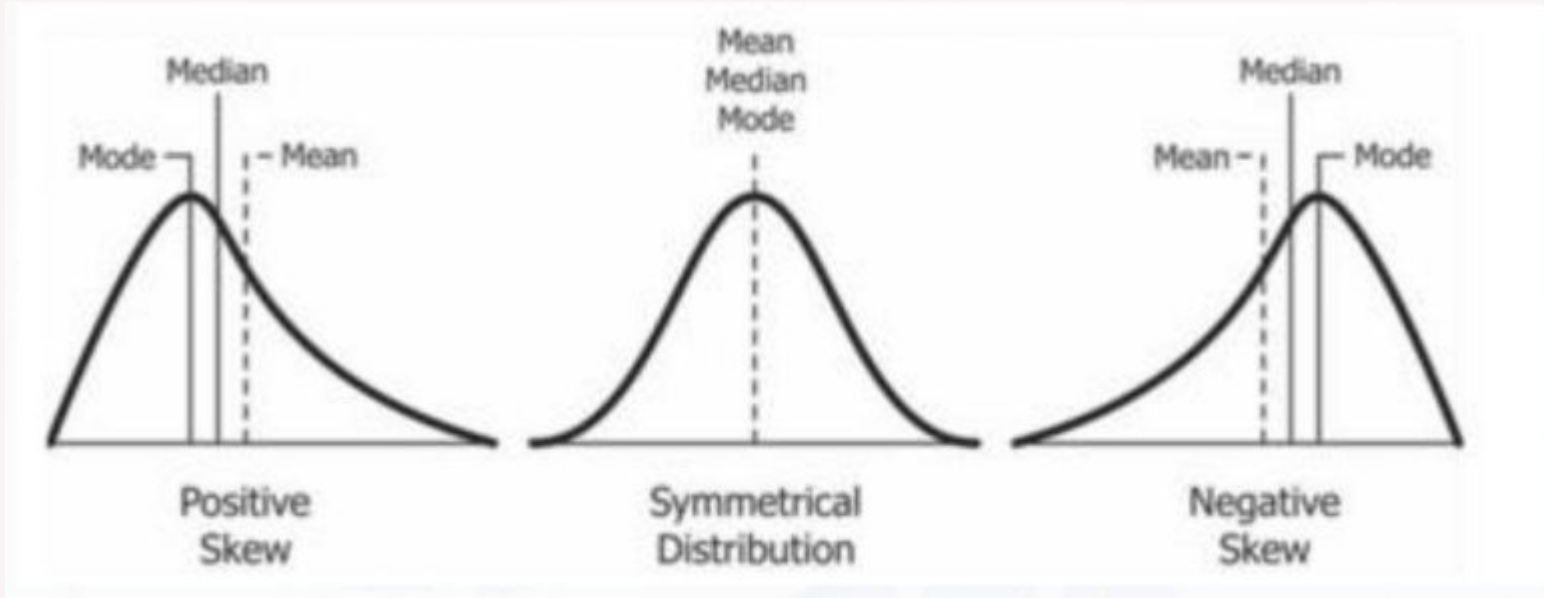
Dispersion Summary

To increase our level of accuracy and sophistication further, it is important that we try to include all the pieces of data in our calculation. This leads us to **variance** and **standard deviation** which are based on the **mean**.

Advantages and disadvantages of measures of dispersion

Measure	Advantages	Disadvantages
Range	• A reasonably good indicator	• Badly affected by extreme values
Inter-quartile range	• Not affected by extreme values • Often used with skewed data	• Does not tell you what happens beyond the quartiles
Variance	• Good measure • All values used • Used when data are fairly symmetrical	• Mathematical properties not useful (SD preferred) • Not so good if data are strongly skewed
Standard deviation	• Good measure • All values used • Used when data are fairly symmetrical • Can be used in mathematical calculations of other statistics	• Not so good if data are strongly skewed

Dispersion Summary

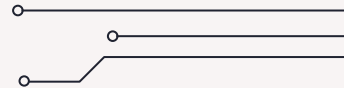


Frequency

Distribution Degree of variation in the data

Can be used for both continuous and categorical data

List, tables or graphs that show the **frequency** of outcomes in a sample.



Frequency

Example: Goals

Sam's team has scored the following numbers of goals in recent games

2, 3, 1, 2, 1, 3, 2, 3, 4, 5, 4, 2, 2, 3

Sam put the numbers in order, then added up:

- how often 1 occurs (2 times),
- how often 2 occurs (5 times),
- etc,

and wrote them down as a **Frequency Distribution** table.

From the table we can see interesting things such as

- getting 2 goals happens most often
- only once did they get 5 goals

Scores:
1,1,2,2,2,2,2,3,3,3,3,4,4,5

Score	Frequency
1	2
2	5
3	4
4	2
5	1

Frequency





Excel Example



Useful Excel Functions

=AVERAGE() Calculates the mean of the selected data

=MEDIAN() Calculates the median of the selected data

=MODE() Calculates the mode of the selected data

=AVERAGEIF() Calculates the average of data IF it satisfies a condition

=STDEV() Calculates the standard deviation of a SAMPLE

=VAR() Calculates the variance of a SAMPLE



Home work

PLEASE SEND YOUR EXCEL SPREADSHEET WITH YOUR ANSWERS THROUGH EMAIL

- Download OR copy & paste the excel dataset from this link:

<https://www.contextures.com/xlSampleData01.html>

- Go straight to the '**Office Supply Sales**' tab and answer the following questions:

1. What was the average price of products purchased in the Central region?
2. Calculate the Mode, Median, and Mean for the first 6 months of purchases.
3. Calculate the Standard Deviation and Variance for the first 6 months of purchases.
4. What does the Standard Deviation tell you in the context of this problem?